SYSTEM ARCHITECTURE & FULL-STACK IMPLEMENTATION SPECIFICATION

# NEPSE Analytics & Trading Intelligence Platform

Target Environment: Antigravity Full-Stack Build Engine

Market: Nepal Stock Exchange (NEPSE) | Timezone: NPT (UTC+5:45)

| TRADING WINDOW | MARKET UNIVERSE | BROKER NETWORK | SETTLEMENT & TAX |
|---|---|---|---|
| **Sun – Thu** | **280+ Scrips** | **~90 Brokers** | **T+2 Rolling** |
| 10:30 Pre \| 11:00–15:00 Cont. | 4 Main + 13 Sub-Indices | Tick-by-Tick Floorsheet ID | 7.5% ST / 5% LT CGT |

## 1. Executive System Scope & Market Mechanics

MODULE 01 // CORE RULES

This blueprint specifies the complete engineering architecture for a production-grade financial analytics web application tailored to the **Nepal Stock Exchange (NEPSE)**. Unlike Western equity platforms, a NEPSE platform must handle Nepali Bikram Sambat (BS) fiscal quarters, dual share classes (Ordinary vs. Promoter shares), book-closure price adjustments for bonus and right shares, Nepal Rastra Bank (NRB) liquidity metrics, and tick-level broker floorsheet tracking.

- **Regulatory Hierarchy: SEBON** (apex regulator for IPOs, mutual funds, and brokers), **NEPSE** (operates NOTS & TMS matching engines), **CDSC** (operates MeroShare & T+2 clearing), **NRB** (regulates Commercial, Development, Finance, and Microfinance institutions), **NIA** (regulates Life/Non-Life insurers), and **ERC** (regulates Hydropower).
- **Session & Circuit Breaker Logic:** Pre-Open runs **10:30–10:45 NPT** (±2% band). Continuous session runs **11:00–15:00 NPT** (±10% daily limit, max ±2% per consecutive tick). Index circuit breakers trigger at **±4%** in the 1st hour (20-min halt), **±5%** in the first 2 hours (40-min halt), and **±6%** anytime (full-day halt).
- **Transaction Cost & WACC Engine:** Portfolio trackers must compute exact net proceeds: Brokerage commission (tiered **0.24%–0.36%** by volume), SEBON fee (**0.015%**), DP charge (flat **NPR 25** per scrip per day per direction), and Capital Gains Tax (**7.5%** individual short-term < 365 days; **5.0%** long-term ≥ 365 days; **10.0%** institutional; **5.0%** dividend tax).

## 2. NEPSE Index & Sub-Index Taxonomy

MODULE 02 // MARKET STRUCTURE

| Index Group | Specific Index Identifiers | Weighting Methodology | Primary Analytical Drivers |
|---|---|---|---|
| **Broad Market** | `NEPSE` (Main), `FLOAT` | Total Market Cap vs. Free-Float Public Shares | Overall market direction; `FLOAT` removes locked-in promoter share distortion. |
| **Blue-Chip Tier** | `SENSITIVE`, `SENSFLOAT` | Class "A" Companies (Total vs. Public Float) | Institutional sentiment; tracks high-capital, profitable, dividend-paying firms. |
| **BFI Sectors (4)** | `BANKING`, `DEVBANK`, `FINANCE`, `MICROFIN` | Sectoral Market Capitalization (~40% of NEPSE) | NRB Monetary Policy, CD Ratio, Base Rate, Interest Spread, NPL %, Distributable Profit. |
| **Insurance (2)** | `LIFEINSU`, `NONLIFE` | Sectoral Market Capitalization | Actuarial valuation, Life Fund growth, Claim ratio, NIA paid-up capital mandates. |
| **Hydropower (1)** | `HYDRO` (90+ Listed Scrips) | Sectoral Market Capitalization | Cost per MW, wet/dry PPA rates, debt service, 3-year promoter/local lock-in expiry. |
| **Real & Services (5)** | `MANUFAC`, `HOTEL`, `TRADING`, `INVEST`, `OTHERS` | Sectoral Market Capitalization | Capacity utilization, tourist arrivals, telecom/reinsurance cash flows (NTC, NRIC). |
| **Collective (1)** | `MUTUAL` (Closed-End Funds) | Fund Unit Capitalization | Weekly/Monthly NAV discount-to-market price, institutional equity |

## 3. End-to-End System Architecture & Tech Stack

MODULE 03 // INFRASTRUCTURE

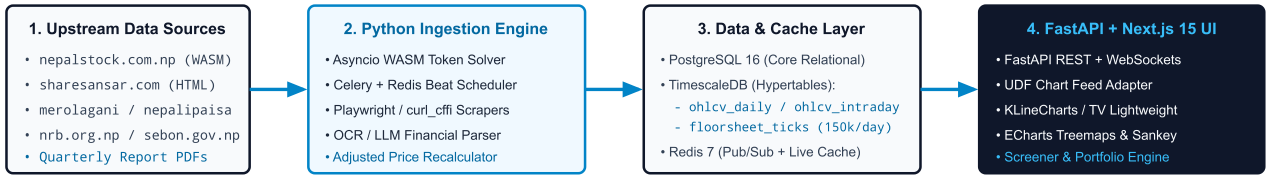



**Figure 3.1: Decoupled 4-Tier Architecture preventing upstream rate-limiting via Redis caching and TimescaleDB hypertables.**

## 4. Reverse-Engineering the Official NEPSE WASM Token

MODULE 04 // SCRAPER CORE

The official portal ( `nepalstock.com.np` ) is an Angular SPA backed by JSON REST endpoints under `/api/nots/...` . Every request requires a dynamic authorization header ( `Authorization: Salter <access_token>` ) generated via a compiled WebAssembly binary ( `cdsc.wasm` ). Do not scrape NEPSE with a slow headless browser; implement the pure-Python WASM token solver:

### Step 1: WASM Handshake & Salt Extraction

- Send `GET /api/authenticate/prove` with realistic browser headers ( `User-Agent` , `Referer: https://nepalstock.com.np/` ).
- The JSON payload returns `serverTime` , `salt1` , `salt2` , `salt3` , `salt4` , `salt5` , `accessToken` , and `refreshToken` .
- Load `cdsc.wasm` once into memory using the Python `wasmtime` or `wasmer` runtime. It exposes five exported integer functions: `cdx(s1, s2, s3, s4, s5)` , `rdx(...)` , `bdx(...)` , `ndx(...)` , and `mdx(...)` .

### Step 2: Token De-obfuscation & POST Payloads

- Pass the 5 salts into the exported WASM functions to compute the exact character indices where dummy digits were injected into `accessToken` .
- Slice out those 5 indices to obtain the valid `valid_access_token` . Cache this token in Redis for **45 seconds** across all workers.
- For `POST` endpoints (e.g., `/api/nots/nepse-data/floorsheet` ), call `/api/nots/nepse-data/market-open` to get the daily `id` , then compute the dynamic POST body `{"id": computed_dummy_id}` from the static array of 100 integers embedded in NEPSE's `main.js` bundle.

> **Critical Upstream Protection Rule (Zero Direct Client Calls)**
> Never allow user browser clients to query NEPSE, ShareSansar, or MeroLagani directly. A single background Celery worker polls NEPSE every **20 seconds** during trading hours (Sun–Thu, 10:30–15:05 NPT), updates Redis hashes ( `nepse:live:quotes` , `nepse:live:indices` ), and broadcasts diffs over FastAPI WebSockets to 10,000+ connected users simultaneously.

## 5. Complete Upstream Scraping & Ingestion Matrix

MODULE 05 // DATA PIPELINES

| Target Source | Key Endpoints / URLs | Extracted Entities | Schedule & Parser Stack |
|---|---|---|---|
| **NEPSE Official** `nepalstock.com.np` | `/api/nots/nepse-data/today-price` `/api/nots/nepse-index` `/api/nots/nepse-data/marketdepth/{id}` `/api/nots/nepse-data/floorsheet` | Real-time LTP, High, Low, Close, Volume, Turnover, 17 Indices, Top 5 Buy/Sell Order Book, Raw Floorsheet ticks. | **Every 20s** (Live Quotes/Indices) **15:15 NPT** (Full Floorsheet) Stack: `httpx` + `wasmtime` |
| **ShareSansar** `sharesansar.com` | `/today-share-price` `/floorsheet` `/company/{symbol}` `/proposed-dividend` | 15-year historical OHLCV backfill, dividend history (Bonus %, Cash %), Book Closure dates, Quarterly Report scans. | **15:30 NPT Daily** (EOD Cross-Check) **Hourly** (Announcements) Stack: `BeautifulSoup4` + `lxml` |
| **MeroLagani** `merolagani.com` | `/CompanyDetail.aspx?symbol={sym}` `/LatestMarket.aspx` `/Ipo.aspx` | Fundamental snapshot verification (EPS, P/E, Net Worth, 52W High/Low), Promoter/Public ratio, news articles. | **Daily 16:00 NPT** Handles ASP.NET `__VIEWSTATE` Stack: `httpx` + `selectolax` |
| **NepaliPaisa** `nepalipaisa.com` | `/api/GetCompanyFundamentals` `/api/GetDividendRightHistory` `/api/GetMutualFundNavs` | Clean structured JSON for historical quarterly balance sheets, P&L statements, and Mutual Fund weekly/monthly NAVs. | **Daily 17:00 NPT** Direct JSON REST consumption Stack: `httpx` AsyncClient |
| **NepseAlpha** `nepsealpha.com` | `/trading-view/history?symbol={s}` `/nepse-data/broker-analysis` | Historical Adjusted OHLCV verification series, Mutual Fund equity holding tables. | **One-time Seed + Weekly Sync** Cloudflare TLS fingerprint bypass Stack: `curl_cffi` (impersonate Chrome) |
| **Nepal Rastra Bank** `nrb.org.np` | Homepage Short-Term Indicators Table Monthly CMEFS `.xlsx` Reports | Daily CD Ratio (%), Interbank Rate (%), Total Deposits, Total Lending, Excess Liquidity, Base Rate, FX Reserves. | **Daily 18:00 NPT** HTML Table + Excel Parser Stack: `BeautifulSoup4` + `openpyxl` |
| **SEBON & CDSC** `sebon.gov.np` / `cdsc` | `/ipo-pipeline`, `/right-share-pipeline` `cdsc.com.np` homepage counters | IPO/FPO/Right Share approval pipeline, issue managers, total Demat & MeroShare user counts. | **Daily 19:00 NPT** Static HTML + PDF table parser Stack: `pdfplumber` + `lxml` |

## 6. PostgreSQL 16 + TimescaleDB Database Schema

MODULE 06 // DATA MODEL

Use a single PostgreSQL 16 instance with the `timescaledb` extension enabled. Relational metadata lives in standard tables; high-volume time-series data lives in partitioned TimescaleDB hypertables with automated continuous aggregates.

### Core Relational Tables (PostgreSQL)

- `companies`: `id` (PK), `symbol` (VARCHAR UNIQUE, e.g., `NABIL`), `nepse_id` (INT), `name`, `sector_id` (FK), `instrument_type` (`EQUITY`, `PROMOTER`, `MUTUAL_FUND`, `DEBENTURE`), `listed_shares` (BIGINT), `public_shares`

### Time-Series Hypertables (TimescaleDB)

- `ohlcv_daily` (Hypertable on `trade_date`): `trade_date` (DATE), `company_id` (INT), `open`, `high`, `low`, `close`, `prev_close`, `volume` (BIGINT), `turnover` (NUMERIC), `transactions` (INT), `adj_factor` (NUMERIC DEFAULT 1.0)

## 7. Sector-Specific Fundamental & Valuation Engine

MODULE 07 // FUNDAMENTALS

All financial statements in Nepal follow the Bikram Sambat (BS) fiscal calendar: **Q1** (Shrawan–Ashwin / Mid-Jul to Mid-Oct), **Q2** (Kartik–Poush / Mid-Oct to Mid-Jan), **Q3** (Magh–Chaitra / Mid-Jan to Mid-Apr), and **Q4** (Baisakh–Ashadh / Mid-Apr to Mid-Jul). Store common metrics in top-level columns of `quarterly_financials` and validate sector-specific metrics in the `sector_metrics` JSONB column via Pydantic schemas:

> **Book Closure Price Adjustment Formula (Bonus & Right Shares):**
> When a stock reaches its `book_closure_date` with $B\%$ Bonus Shares and $R\%$ Right Shares (where Par Value = NPR 100):
> $P_{\text{adjusted}} = (P_{\text{market}} + 100 \times R) \div (1 + B + R)$ | Historical Multiplier Factor: $F_{\text{adj}} = P_{\text{adjusted}} \div P_{\text{market}}$
> *Application:* Multiply all historical `open`, `high`, `low`, `close` prices prior to `book_closure_date` by cumulative $F_{\text{adj}}$ when the user activates the `ADJ` chart toggle.

| Sector Group | Mandatory JSONB Schema Keys (`sector_metrics`) | Automated Valuation & Health Flags |
|---|---|---|
| **BFIs** (Comm. Banks, Dev. Banks, Finance, Microfinance) | `npl_pct` (Non-Performing Loans), `cost_of_funds_pct`, `base_rate_pct`, `interest_spread_pct`, `cd_ratio_pct`, `car_pct` (Capital Adequacy), `distributable_eps`, `loan_loss_provision`. | **Dividend Capacity Estimator:** `distributable_profit / paid_up_capital`. Flag red if `npl_pct > 4.5%` or `car_pct < 11%` (NRB cap). Graham Number = $\sqrt{(22.5 \times EPS \times BVPS)}$. |
| **Hydropower** (90+ Projects) | `installed_capacity_mw`, `total_project_cost`, `cost_per_mw`, `debt_to_equity`, `ppa_wet_rate` (NPR 4.80), `ppa_dry_rate` (NPR 8.40), `cod_date`, `promoter_lockin_expiry`, `royalty_expense`. | **Cost/MW Benchmark:** Flag green if `cost_per_mw < NPR 180M`. Display high-visibility countdown badge 90 days before `promoter_lockin_expiry` (3 years post-IPO allotment). |
| **Life & Non-Life Insurance** | `life_fund_amount`, `gross_premium`, `net_premium`, `claim_settlement_ratio`, `solvency_margin`, `unearned_premium_reserve`, `investment_income`, `actuarial_bonus_rate`. | **Price-to-Life-Fund Ratio:** Compare Market Cap against `(paid_up_capital + reserves + 0.15 × life_fund_amount)`. Track NIA minimum paid-up capital compliance (NPR 5B Life / 2.5B Non-Life). |
| **Manufacturing, Hotels & Trading** | `operating_margin_pct`, `ebitda`, `inventory_turnover`, `receivables_days`, `capacity_utilization_pct`, `free_cash_flow`. | **EV/EBITDA & DCF Model:** Enterprise Value = $Market\ Cap + Total\ Debt - Cash$. Automated 5-year DCF calculator using NRB 10-yr bond yield as risk-free rate. |
| **Mutual Funds** | `weekly_nav`, `monthly_nav`, `equity_holding_pct`, `fixed_deposit_pct`, `cash_balance_pct`, `maturity_date`, `top_10_holdings` (array). | **NAV Discount Screener:** $Discount\ \% = ((LTP - Weekly\ NAV) \div Weekly\ NAV) \times 100$. Highlight funds trading at >15% discount within 12 months of maturity. |

## 8. Interactive Trading Chart & Technical Suite

MODULE 08 // CHARTING ENGINE

Embed **KLineCharts Pro (v9+)** or **TradingView Lightweight Charts™** inside the Next.js workspace, backed by a TradingView UDF-compatible FastAPI router so you can also drop in the private TradingView Advanced Charting Library seamlessly:

### Charting Capabilities & UDF API Contract

- **UDF Endpoints:** Implement `GET /api/udf/config`, `GET /api/udf/symbols?symbol=NABIL`, `GET /api/udf/search?query=`, and `GET /api/udf/history?symbol=NABIL&from=...&to=...&resolution=1D&adjusted=true`

### Custom NEPSE Chart Overlays (Differentiators)

- **1-Click `ADJ` / `UNADJ` Toggle:** Instantly switches between raw historical exchange prices and

## 9. "Smart Money" Floorsheet & Broker Intelligence

MODULE 09 // FLOORSHEET ENGINE

Every single trade on NEPSE discloses the exact `buyer_broker` and `seller_broker` IDs. Because institutional funds, high-net-worth operators, and mutual funds execute through specific brokers, quantitative floorsheet analysis is the highest-demand retail feature in Nepal.

**Core Broker Quantitative Formulas (Computed Over Rolling Window $W \in \{1D, 5D, 15D, 30D, 90D\}$):**

1. **Broker Accumulation Ratio ($BAR_{b,s}$):** $(BuyQty_{b,s} - SellQty_{b,s}) \div TotalMarketVolume_{s}$
2. **Broker Buy VWAP ($VWAP_{b,s}$):** $\sum(BuyQty_{b,s,i} \times Rate_{i}) \div \sum BuyQty_{b,s,i}$ (Tracks exact operator cost basis; $Operator\ Margin\ \% = (LTP - VWAP_{b,s}) \div VWAP_{b,s}$)
3. **Top-3 Broker Concentration (Cornering Index $CI_s$):** $\sum_{k=1..3} NetBuyQty_{k,s} \div TotalMarketVolume_{s}$ (Alert if $CI_s > 0.45$ and Float Cap < NPR 1.5B)

### Floorsheet Analytics Features to Build

- **Smart Money Accumulation Scanner:** Ranks all 280+ scrips where the top 5 buyers have accumulated >35% of total traded volume over 10 consecutive trading days while price volatility remained compressed (<4% ATR).
- **Operator Cost-Basis Matrix:** For any selected stock (e.g., `SHPC`), displays a sorted table of all 90 brokers showing Buy Qty, Sell Qty, Net Holding Change, Buy VWAP, Sell VWAP, and whether the top holding brokers are currently in unrealized profit or loss.
- **Broker-to-Broker Sankey Flow Diagram:** Apache ECharts Sankey visualization showing exact share transfer pipelines (e.g., Broker #42 & #14 distributing directly into Broker #58 & #45).

### Intraday Tape & Anomaly Detection

- **Whale / Block Trade Alert Stream:** Live WebSocket feed flagging single floorsheet ticks where `quantity ≥ 5,000` shares or `amount ≥ NPR 2,500,000`.
- **Self-Trade / Matching Wash Filter:** Identifies trades where `buyer_broker == seller_broker` (matching transactions) to separate genuine accumulation from artificial volume churning.
- **Broker Profile Deep-Dive ( `/broker/58` ):** Dedicated dashboard for each broker number showing their top 10 accumulated stocks today, this week, and this month, plus sector rotation preferences.

## 10. Multi-Factor Screener, Macro & Portfolio Suite

MODULE 10 // USER TOOLS

| Application Module | Technical Specification & Query Logic | Frontend Visualization Components |
|---|---|---|
| **Dynamic Stock Screener** (Technical + Fundamental + Floorsheet) | AST-based JSON filter builder compiling into safe parameterized SQL against a denormalized materialized view `mv_screener_snapshot` refreshed every 1 min. Supports cross-domain queries: e.g., `sector = 'HYDRO' AND rsi_14 < 35 AND close > sma_200 AND top3_broker_accum_15d_pct > 30 AND cost_per_mw < 200000000`. | Filter pill builder, preset one-click strategies (*"Undervalued Banks"*, *"Broker Cornered Low-Cap Hydro"*, *"RSI Bullish Divergence"*), CSV/Excel export. |
| **NRB Macro & Liquidity Radar** | Stores daily NRB time-series in `macro_indicators`. Computes 30-day rate-of-change on CD Ratio and Interbank Rate to generate an automated **Liquidity Regime Score (0–100)** (Bullish when CD Ratio < 82% and Interbank Rate < 3.5%). | Dual-axis chart overlaying NEPSE Index against Inverse Interbank Rate & CD Ratio; Monetary Policy tracker cards. |
| **Sector Relative Rotation Graph (RRG)** | Calculates JdK RS-Ratio (Relative Strength vs. `NEPSE` index) and JdK RS-Momentum for all 13 sectoral sub-indices over rolling 10-day and 10-week windows. | 4-Quadrant scatter plot (*Leading, Weakening, Lagging, Improving*) with animated trailing tails showing sector rotation. |
| **Market Breadth & Heatmap** | Computes Advance/Decline/Unchanged counts, McClellan Oscillator, % of stocks above 20/50/200 | Interactive ECharts Treemap sized by Market Cap or Turnover, colored by % |

## 11. Monorepo Directory Structure & API Specification

MODULE 11 // CODEBASE LAYOUT

### Monorepo File Tree (Antigravity Target)

- `nepse-platform/`
  - `docker-compose.yml` (Postgres/TimescaleDB, Redis, FastAPI, Celery, Next.js)
  - `backend/` (Python 3.12 + FastAPI + SQLAlchemy 2.0 Async)
    - `app/core/`: `config.py`, `database.py`, `redis.py`, `nepse_calendar.py`
    - `app/scrapers/`: `nepse_wasm.py`, `cdsc.wasm`, `sharesansar.py`, `merolagani.py`, `nrb_macro.py`, `pdf_ocr_parser.py`
    - `app/services/`: `adj_price_engine.py`, `broker_analytics.py`, `technical_engine.py`, `wacc_tax_calc.py`
    - `app/api/v1/`: `market.py`, `udf_chart.py`, `floorsheet.py`, `screener.py`, `fundamentals.py`, `portfolio.py`, `ws.py`
    - `app/workers/`: `celery_app.py`, `live_poller.py`, `eod_pipeline.py`
  - `frontend/` (Next.js 15 App Router + TypeScript + TailwindCSS + shadcn/ui)
    - `src/app/`: `(dashboard)/page.tsx`, `company/[symbol]/page.tsx`, `chart/page.tsx`, `floorsheet/page.tsx`, `screener/page.tsx`, `broker/[id]/page.tsx`, `macro/page.tsx`, `portfolio/page.tsx`
    - `src/components/chart/`: `NepseTradingChart.tsx`, `BrokerVwapOverlay.ts`

### Core FastAPI REST & WebSocket Endpoints

- `GET /api/v1/market/summary`: Live NEPSE index, 13 sub-indices, turnover, advance/decline breadth, market status (`PRE_OPEN`, `OPEN`, `HALTED`, `CLOSED`), and top gainers/losers/turnover.
- `GET /api/v1/company/{symbol}/full`: Profile, live quote, market depth, 5-year quarterly financials, sector metrics JSONB, dividend history, and lock-in countdown.
- `GET /api/v1/udf/history`: TradingView/KLineCharts historical candle endpoint supporting `symbol`, `resolution` (`1` .. `1D`), `from`, `to`, and `adjusted=true|false`.
- `GET /api/v1/floorsheet/analytics`: Accepts `symbol`, `broker_no`, `days` (1..90). Returns Broker Accumulation Ratio, Buy/Sell VWAP, Cornering Index, and Sankey matrix.
- `POST /api/v1/screener/run`: Executes multi-condition technical, fundamental, and broker accumulation filter payload against `mv_screener_snapshot`.
- `WS /api/v1/ws/live`: Multiplexed WebSocket pushing 20-second delta frames for index ticks, scrip LTP changes, market depth, and whale floorsheet alerts.

## 12. Step-by-Step Build & Execution Roadmap

MODULE 12 // IMPLEMENTATION PLAN

| Sprint Phase | Core Deliverables | Key Implementation Tasks | Verification & Acceptance Criteria |
|---|---|---|---|
| **Phase 1** Foundation & DB | Docker stack, PostgreSQL + TimescaleDB schema, BS/AD date converter. | Initialize migrations for `companies`, `ohlcv_daily`, `floorsheet_ticks` hypertables, and continuous aggregates. | Hypertable chunk compression and sub-50ms query on 10M synthetic floorsheet rows. |
| **Phase 2** WASM Scraper & Backfill | `nepse_wasm.py` live worker + ShareSansar/NepaliPaisa historical seed scripts. | Implement `cdsc.wasm` token generator; backfill 10 years of daily OHLCV, dividends, and 90 days of floorsheets. | Zero HTTP 401/403 errors from NEPSE; Redis cache updates every 20s during market hours. |
| **Phase 3** Charting & Adj... | Book-closure adjustment math, UDF API, KLineCharts (TV Lightweight... | Compute cumulative `adj_factor` f... | Toggling `ADJ` smoothly ... |